

BobaLink Browser Extension “ Changelog

Revision: 1 **Last modified:** 2026-06-10T23:55:00Z **Scope:** BobaLink (extension/) “ WXT + TypeScript Manifest-V3 cross-browser extension that detects magnet links and .torrent URLs and forwards them to the Boba merge service on port 7187. **Authority:** docs/browser_extension/Status.md (Rev 7), docs/browser_extension/RELEASE_READINESS.md, git log --oneline.

Format: [Keep a Changelog](#), [Semantic Versioning](#). **Anti-bluff** (11.4.6 / 11.4.123). Every entry cites a real commit hash from the Session-11 commit set (024210f 5edf6ac 2011810 5fa78d9 e80c9d9 5e44c85 750e4e5, verified via git log --oneline). The five Fixed defects each trace to the Status.md Rev 7 / RELEASE_READINESS.md 3 record. No invented entries.

[1.0.0] “ Unreleased

Status: code-complete + comprehensively tested, NOT yet released. Per RELEASE_READINESS.md 6, a store release is gated on (a) the live-backend detect “ send “ torrent-in-qBittorrent round-trip against a running :7187 turning GREEN (currently INFRA-BLOCKED, honest SKIP “ fetchImpl stub substitutes the network boundary) and (b) operator-produced store assets (screenshots, promotional images, listing submission). Phases 1“3 are PASS; Phases 4“9 are IN-PROGRESS with enumerated PENDING items.

Added

- **WXT build wiring “ loadable cross-browser artifacts.** WXT entrypoints at src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html} (thin wrappers over the existing logic modules, so all prior test imports stay valid); wxt build produces a loadable .output/chrome-mv3/ (manifest_version:3, service worker, popup, options, content-script, icons 16/32/48/128) AND .output/firefox-mv2/. Content-script matches are derived at build time from the curated SITE_SELECTORS table “ no <all_urls>. (024210f, 5edf6ac)
- **Tab-group batched send (MENU_SEND_GROUP).** src/tabgroups/index.ts dedupes detected torrents across a tab group (infohash-first) and dispatches the batch as ONE addMagnets POST, wired into background/index.ts's MENU_SEND_GROUP context-menu action. Manifest gains the minimal-privilege tabGroups permission only (research-confirmed tabs not needed “ only tab.id is read). Group-send hardening: offline-queue enqueue-on-failure, network-error notification, hardened async flush. (5edf6ac, 2011810)
- **Phase-7 decrypt-before-send credential path.** BobaClient.create() decrypts the ServerConfig.encryptedBobaApiToken AES-256-GCM bundle via shared/crypto; the background reads the session passphrase from chrome.storage.session (in-memory, never disk-persisted) and sends the decrypted PLAINTEXT token as a bearer header. Default-open when locked / no token “ the ciphertext is never sent. The token value and passphrase are never logged. (e8fde43, 2011810)
- **8-locale internationalization.** _locales/{en,ru,de,fr,es,it,pt,ja} “ the plan's 8-locale target reached “ each at 29-key + placeholder parity, packaged into both

the chrome and firefox builds, mutation-proven by `locale-parity.test.ts` (every non-en locale checked against en). (5fa78d9, e80c9d9, 750e4e5)

- **Accessibility â€” WAI-ARIA structure + keyboard navigation.** Structural ally tests (`tests/ally/{popup,options}.ally.test.ts`, 18 â€” roles / accessible names / tablistâ€”i,Žtabpanel / live-regions) plus the options tablist WAI-ARIA Arrow/Home/End keyboard pattern (`tests/ally/keyboard-nav.ally.test.ts`, 18), all mutation-proven. (2011810, 5e44c85)
- **Multi-type test corpus.** Performance/benchmark (`tests/perf/* â€” parser / scanner / crypto distributions, 10—regression-catching, 3—deterministic`), stress (`tests/stress/{parsers,queue}.stress.test.ts â€” â‰¥1000 enqueue, FIFO, concurrent`), chaos (`tests/chaos/queue.chaos.test.ts â€” corruption, soft/hard fail, dead-letter, send-failure injection`), security (`tests/security/*`), and integration (`tests/integration/pipeline.test.ts â€” detect â†’ POST : 7187 request URL+body â†’ queue persist+drain over the real sendMessageâ†’ onMessage bridge`). (fa03323, 024210f, 2011810, 5e44c85, 750e4e5)
- **Manual CI gate extension/ci-ext.sh.** tsc â†’ lint â†’ full Vitest â†’ chrome + firefox builds â†’ Â§11.4.38 artifact-verify (opens the produced `manifest.json` and asserts every referenced asset + the default_locale â†’ _locales/ catalog exist non-zero on disk) â†’ per-store wxt zip (boba-link-1.0.0-{chrome,firefox,sources}.zip). Final line CI-EXT: PASS. No CI/CD â€” manual only. (5fa78d9)
- **Challenges + HelixQA bank.** challenges/extension/detect_and_forward_challenge.sh drives the REAL orchestrator + client end-to-end (PASS on captured evidence, mutation-verified: no-op stub â†’ FAIL); plus decrypt_and_send_challenge.sh, offline_queue_recovery_challenge.sh; HelixQA bank boba-boba-link.yaml (BOBA-LINK-007/008/009/010, http: cases live-gated on : 7187). (2011810, 5e44c85)

Fixed

- **_locales packaging â€” would have FAILED Chrome load.** The manifest declared `default_locale: "en" + __MSG_*__` but the build packaged no `_locales/` (WXT copies static assets from `src/public/`, the catalog lived at `src/_locales/`), so Chrome would have rejected the unpacked extension. Fixed by moving the catalog to `src/public/_locales/`; permanent guard is the `ci-ext.sh` Â§11.4.38 artifact-open step. (5fa78d9)
- **Popup false-failure â€” popupâ€”i,Žbackground contract bug.** The popup read `r.torrent.id` but the real background returns a flat `SendOutcome {id, â€”}`, so a SUCCESSFUL send threw in the popup and showed a false "Send failed" (the row never flipped to Sent). Fixed (popup reads `r.id`; the popup-unit fake that had masked it corrected; integration assertion reconciled per Â§11.4.120). (e80c9d9)
- **Scan-result trust-overwrite â€” content-script trust bug.** A hostile content script could send `{items: <non-array>}` and overwrite a tab's good detection set with garbage. Fixed with the `isValidScanResult` shape-guard in the background message router (the non-array test seeds-goodâ†’corrupt and asserts the good set survives). (e80c9d9)
- **ReDoS in sanitizedDisplayName â€” O(nÂ²) â†’ linear.** A hostile magnet dn of `'<^—100k took 4s+ and could hang the content script. Fixed /<[^>]*>/g â†’ /<[^<>]*>/g: linear time (output identical on valid input, strictly safer on hostile â€” now catches <scr<script>ipt> evasion). 19 hostile-input tests in tests/security/scanner-hostile-input.test.ts. (5e44c85, src/parser/magnet.ts)`
- **Ally keyboard gap â€” WAI-ARIA tablist arrow-key pattern (WCAG 2.1.1).** The options tablist lacked the arrow-key keyboard pattern; keyboard-only users could not

traverse tabs. Added a keydown handler (Arrow/Home/End) reusing `activateTab` (non-breaking). 18 keyboard-nav tests. (5e44c85, `src/options/options.ts`)

Security

- **ReDoS hardening of the magnet display-name sanitizer** (see Fixed above) â€” linear-time tag stripping, hostile-input fuzz corpus. (5e44c85)
 - **Scan-result trust-guard** â€” `isValidScanResult` shape-validation at the contentâ†’background trust boundary so a hostile content script cannot poison a tab's detection set (16 message-router-robustness tests). (e80c9d9)
 - **Least-privilege manifest.** No `<all_urls>` (content-script matches derived from `SITE_SELECTORS`); no `scripting/tabs/cookies`; `tabGroups` added only for the Phase-5 batch send; `host_permissions = http://localhost:7187/*` only; CSP `script-src 'self'`. Backed by `tests/security/*` (least-privilege manifest, CSP, no-hardcoded-secret, secret-storage, content-XSS) and the Â§11.4.10.A challenges/`security/credential_leak_audit.sh` (mutation-verified). (024210f, 5fa78d9)
 - **No embedded key.** The Boba token is stored AES-256-GCM-encrypted under a user-supplied session passphrase (`chrome.storage.session`, never disk); the plaintext token and passphrase are never logged or persisted. (2011810)
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Sources verified 2026-06-10

- `git log --oneline` â€” Session-11 commits 024210f 5edf6ac 2011810 5fa78d9 e80c9d9 5e44c85 750e4e5
- `docs/browser_extension/Status.md` (Rev 7) â€” per-phase status, baseline facts, Â§11.4.138 bluff-audit
- `docs/browser_extension/RELEASE_READINESS.md` (Rev 1) â€” five defects, release-blocker enumeration, honest classification
- `extension/src/**` â€” the real modules each entry traces to (`background/index.ts`, `content/index.ts`, `api/boba-client.ts`, `parser/magnet.ts`, `options/options.ts`, `tabgroups/index.ts`, `entrypoints/*`)